

# 2009 MI PU3 H2 Maths Prelims 2 Paper 2 Solutions

| QN                     | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>1</b><br><b>[5]</b> | <p>Graph of <math>y = \frac{2}{x}</math> with two ends not touching the axes</p> <p>Volume required = <math>\frac{1}{3}\pi(1)^2(2) + \pi \int_2^4 \left(\frac{2}{y}\right)^2 dy</math></p> $= \frac{2}{3}\pi + 4\pi \left[\frac{y^{-1}}{-1}\right]_2^4$ $= \frac{5}{3}\pi \text{ or } 1\frac{2}{3}\pi \text{ units}^3$                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>2</b><br><b>[7]</b> | $\frac{d^2y}{dx^2} = -e^{-y} \frac{dy}{dx}$ $\frac{d^3y}{dx^3} = -e^{-y} \frac{d^2y}{dx^2} + e^{-y} \left(\frac{dy}{dx}\right)^2 = e^{-y} \left[\left(\frac{dy}{dx}\right)^2 - \frac{d^2y}{dx^2}\right] \text{ (shown)}$ $\frac{d^4y}{dx^4} = e^{-y} \left[3 \frac{dy}{dx} \frac{d^2y}{dx^2} - \frac{d^3y}{dx^3} - \left(\frac{dy}{dx}\right)^3\right] \text{ or equivalent}$ <p><math>x = 0, y = 0: \frac{dy}{dx} = -1, \frac{d^2y}{dx^2} = 1, \frac{d^3y}{dx^3} = 0, \frac{d^4y}{dx^4} = -2 \text{ (shown)}</math></p> $y = -x + \frac{1}{2}x^2 - \frac{1}{12}x^4 + \dots$ $e^x - 1 - 2x - \frac{1}{6}x^3 - \frac{1}{8}x^4$ $= \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4\right) - 1 - 2x - \frac{1}{6}x^3 - \frac{1}{8}x^4$ $= y \text{ (verified)}$ |
| <b>3</b><br><b>[7]</b> | <p>(i) Area of R = Area of trapezium (or triangle + rectangle) - <math>\int_1^{\sqrt{2}} \sqrt{\frac{x^2-1}{x^2}} dx</math></p> $= \frac{1}{2}(\sqrt{2} + \sqrt{2} - \sqrt{\frac{1}{2}})\sqrt{\frac{1}{2}} - \int_1^{\sqrt{2}} \sqrt{\frac{x^2-1}{x^2}} dx$ <p>(ii) <math>\frac{dx}{d\theta} = \sec\theta \tan\theta</math></p> <p>when <math>x = \sqrt{2}, \theta = \frac{\pi}{4}</math> and when <math>x = 1, \theta = 0</math></p> $\int_0^{\pi/4} \sqrt{\frac{\sec^2\theta - 1}{\sec^2\theta}} \sec\theta \tan\theta d\theta$ $= \int_0^{\pi/4} \tan^2\theta d\theta$ $= \int_0^{\pi/4} \sec^2\theta - 1 d\theta$ $= 1 - \frac{\pi}{4}$ <p>Area of R = <math>\frac{3}{4} - (1 - \frac{\pi}{4}) = \frac{1}{4}(\pi - 1) \text{ units}^2</math></p>                         |

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| <p><b>4</b><br/><b>[7]</b></p>  | <p>(i) least <math> z  = 1</math></p> <p>Greatest <math> z  = 3</math></p> <p>(ii) <math>z = (1 + 2\cos \frac{\pi}{3}) + i(-2\sin \frac{\pi}{3})</math></p> <p><math>= 2 - \sqrt{3}i</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>5</b><br/><b>[14]</b></p> | <p>(i) <math>f^{-1}: x \rightarrow -\sqrt{3-x}, x \leq 3</math></p> <p>(ii) Graphs of <math>f(x)</math>, <math>f^{-1}(x)</math> and <math>ff^{-1}</math></p> <p>(iii) 2 methods: either by solving graphically of <math>-x^2 + 3 = -\sqrt{3-x}</math> or <math>-(x^2 - 3)^2 = 3 - x</math> or solving by formula for <math>-x^2 + 3 = x</math>.<br/><math>x = -2.30</math></p> <p>(iv) <math>hf</math> cannot be formed since <math>R_f = (-\infty, 3] \not\subset D_h = (0, \infty)</math></p> <p>(v) <math>hg</math> exists if <math>R_f = (0, 3]</math><br/>Largest <math>T = (-\sqrt{3}, \sqrt{3})</math></p> <p>(vi) <math>hg(x) = h(-x^2 + 3)</math><br/><math>= \ln(3 - x^2), x \in (-\sqrt{3}, \sqrt{3})</math></p> <p><math>R_{hg}</math> is <math>y \leq \ln 3</math> or 1.10</p> |
| <p><b>6</b><br/><b>[4]</b></p>  | <p>Reason: too costly or time consuming to take population or size of population is too big etc.</p> <p>Example should illustrate</p> <ul style="list-style-type: none"> <li>• correct strata size;</li> <li>• random selection from each stratum</li> </ul> <p>Advantage: a good representative sample of the population (or equivalent)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><b>7</b><br/><b>[6]</b></p>  | <p>(a) Number of ways = <math>14 \cdot {}^{13}C_5</math><br/><math>= 5148</math></p> <p>(b) Number of ways = <math>5.3 + 5.2 + 3.2</math><br/><math>= 31</math></p> <p>(c) Number of ways = <math>{}^8C_4 \cdot {}^4C_2 \cdot 1</math> or <math>\frac{8!}{4!2!2!}</math><br/><math>= 420</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| <p><b>8</b><br/><b>[7]</b></p>  | <p>(i) <math>H_0: \mu = 1.5</math><br/> <math>H_1: \mu &gt; 1.5</math><br/> <math>\alpha = 0.05</math><br/> Using GC, <math>\bar{X} = 1.5625</math>, <math>s = 0.1408</math><br/> Since <math>n</math> is small (<math>&lt; 30</math>), use t-test.<br/> <math display="block">T = \frac{\bar{X} - \mu}{s/\sqrt{n}} = 1.2555 \text{ or } p\text{-value} = 0.1248</math> Since <math>p\text{-value} &gt; 0.05</math>, we do not reject <math>H_0</math>.<br/> There is insufficient evidence at 5% level of significance to conclude that the importer is understating the average mass of a durian.<br/> (ii) Necessary to assume population is normally distributed.</p>                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>9</b><br/><b>[8]</b></p>  | <p>(a)(i) <math>X</math> = number of flashlights with a faulty bulb<br/> <math>X \sim B(10, 0.01)</math><br/> <math>P(X = 0) = 0.9044</math> or <math>0.904</math> (3sf)<br/> (ii) <math>P(\text{faulty bulb})P(\text{faulty switch})</math><br/> <math>= (0.01)(0.015)</math><br/> <math>= 0.00015</math><br/> <math>Y</math> = number of flashlights with both a faulty bulb and a faulty switch<br/> <math>Y \sim B(80, 0.00015)</math><br/> <math>P(Y \geq 1) = 1 - P(Y = 0)</math><br/> <math>= 0.0120</math> (3sf)<br/> (b) <math>X</math> = number of children without immunisation<br/> <math>X \sim B(60, 0.07)</math><br/> Since <math>n &gt; 50</math> and <math>np = 4.2 &lt; 5</math>,<br/> <math>X \sim \text{Po}(4.2)</math> approximately<br/> <math>P(X \leq 2) = 0.210</math> (3sf)</p>                                                                                                                                                                                                                                                                    |
| <p><b>10</b><br/><b>[9]</b></p> | <div data-bbox="272 1171 535 1312" data-label="Figure"> </div> <p>(i) <math>r = 0.972</math> (3sf)<br/> Answer of <math>r</math> is close to 1 which suggests a strong, positive linear correlation between <math>x</math> and <math>y</math>.<br/> So linear model is appropriate.</p> <div data-bbox="279 1459 544 1600" data-label="Figure"> </div> <p>(ii)</p> <p>(iii) Equation of regression line of <math>y</math> on <math>x</math> is <math>y = 10.1x + 2.58</math>.<br/> Slope represents expected increase of 10.1 dead cockroaches for every 1% increase in the concentration of the insecticide.</p> <p>(iv) Not reliable to estimate the value of <math>y</math> for larger values of <math>x</math> as the linear relationship between <math>x</math> and <math>y</math> may not hold for values outside of the range of the data.<br/> OR Not reliable as larger values of <math>x</math> are out of range so data may not be linearly correlated.</p> <p>(v) when <math>x = 0</math>, <math>y = 2.58</math>.<br/> Average number of cockroaches = 2.58.</p> |

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| <p><b>11</b><br/><b>[12]</b></p> | <p>(a)(i) <math>P(A) = 18/36 = \frac{1}{2}</math> (or equivalent)<br/> <math>P(B) = 2(5/6)(1/6) + (1/6)(1/6) = 11/36</math> (or equivalent)<br/> <math>P(C) = 3/36 = 1/12</math> (or equivalent)</p> <p>(ii) <math>P(B \cap C) = 2/36 = 1/18</math><br/> <math>P(B)P(C) = 11/432</math><br/> Since <math>P(B \cap C) \neq P(B)P(C)</math>, so B and C are not independent.</p> <p>(iii) <math>P(C'   A) = P(A \cap C') / P(A)</math><br/> <math>= (15/36) \div (1/2) = 5/6</math> (or equivalent)</p> <p>(b)(i) <math>P(\text{call for Daisy and Daisy is at home})</math><br/> <math>= (0.2)(0.75)</math><br/> <math>= 0.15</math></p> <p>(ii) <math>P(\text{person to whom call is made is at home})</math><br/> <math>= 0.15 + (0.3)(0.5) + (0.5)(0.8)</math><br/> <math>= 0.70</math></p>                                                                                               |
| <p><b>12</b><br/><b>[14]</b></p> | <p><math>A \sim N(291, 81)</math>, <math>B \sim N(282, 144)</math></p> <p>(i) <math>P(A &gt; 300)P(B &gt; 300)</math><br/> <math>= P(Z &gt; \frac{300 - 291}{9}) P(Z &gt; \frac{300 - 282}{12})</math><br/> <math>= 0.0106</math> (3sf)</p> <p>(ii) <math>P(550 &lt; A + B &lt; 600)</math><br/> <math>= P(\frac{550 - 573}{15} &lt; Z &lt; \frac{600 - 573}{15})</math><br/> <math>= P(-1.5333 &lt; Z &lt; 1.8)</math><br/> <math>= 0.901</math> (3sf)</p> <p>(iii) <math>P(B &gt; A)</math><br/> <math>= P(B - A &gt; 0)</math><br/> <math>= P(Z &gt; \frac{0 - (-9)}{15})</math><br/> <math>= P(Z &gt; 0.6)</math><br/> <math>= 0.274</math></p> <p><math>P(A \geq m) \leq 0.05</math><br/> <math>P(Z &gt; \frac{m - 291}{9}) \leq 0.05</math><br/> <math>\frac{m - 291}{9} &gt; 1.6449</math><br/> <math>m &gt; 305.8</math><br/> Least <math>m = 306</math> (nearest whole number)</p> |